

Model Building- GAM Models

Load necessary packages

```
library(tidyverse)
library(tidymodels)
library(ggplot2)
library(purrr)
library(caret)
library(glmnet)
library(mgcv)
library(splines)
library(pdp)
library(earth)
```

Load in dataset and use 0/1 for binary variables

```
data <- read.csv("data/data_clean.csv") |>
  data.frame()

data$UnablePregnant <- ifelse(data$UnablePregnant == "Yes", 1,
                              ifelse(data$UnablePregnant == "No", 0, NA))

data$RegularPeriod_12m <- ifelse(data$RegularPeriod_12m == "Yes", 1,
                                  ifelse(data$RegularPeriod_12m == "No", 0, NA))

data = data |>
  mutate(
    UnablePregnant = as.factor(UnablePregnant),
    RegularPeriod_12m = as.factor(RegularPeriod_12m)
  )

data_under50 = data |>
  filter(Age < 50)
```

All participants

Model using outcome as Age_FirstPeriod (counts)

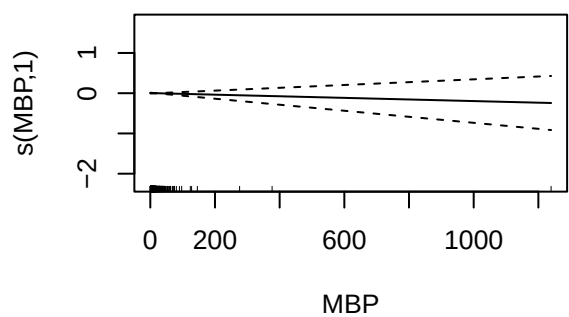
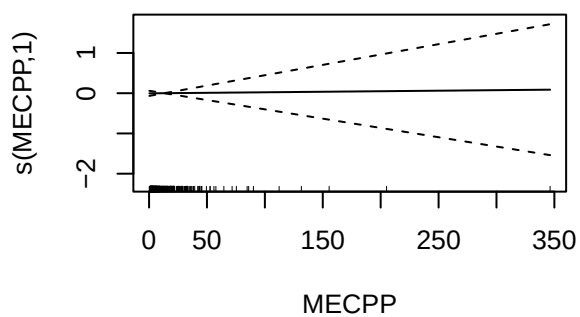
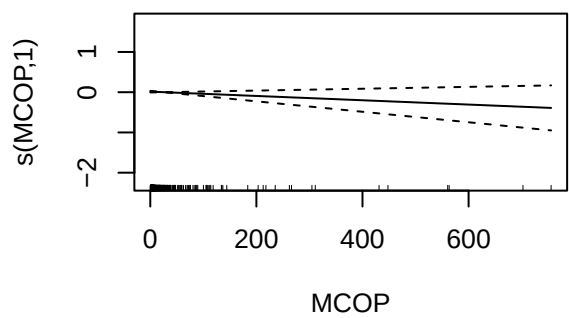
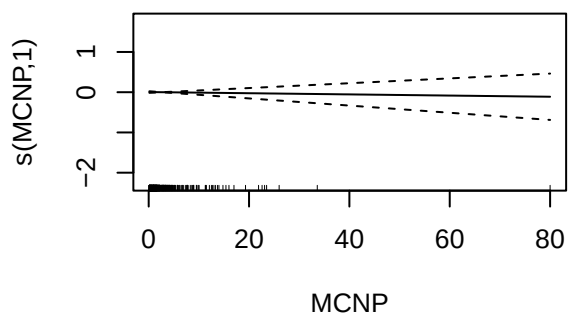
```
# Model Fitting using mgcv
```

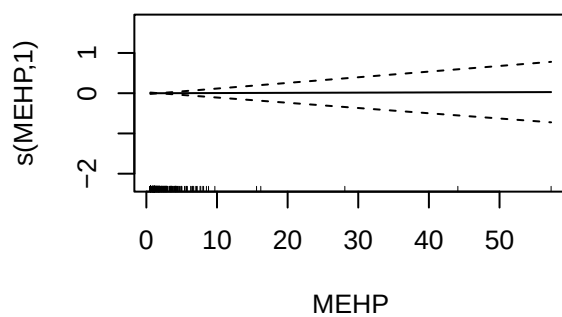
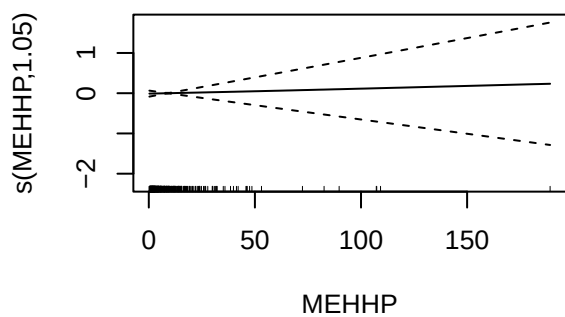
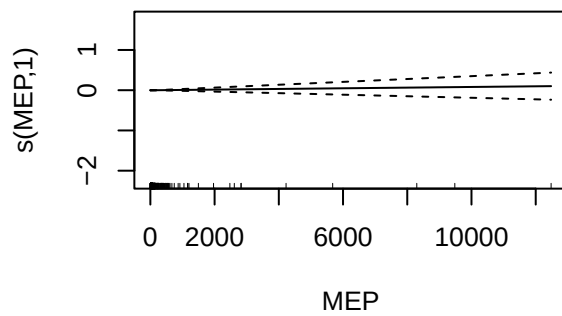
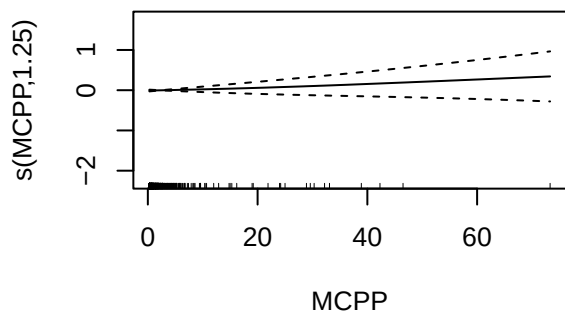
```
# Fit model
```

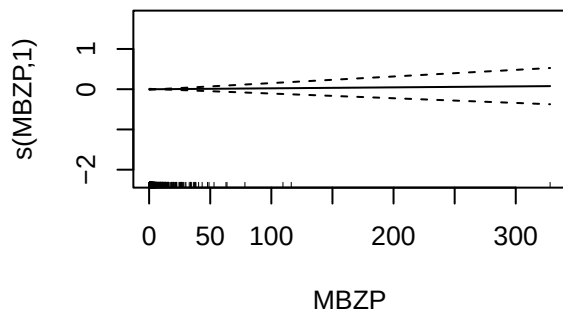
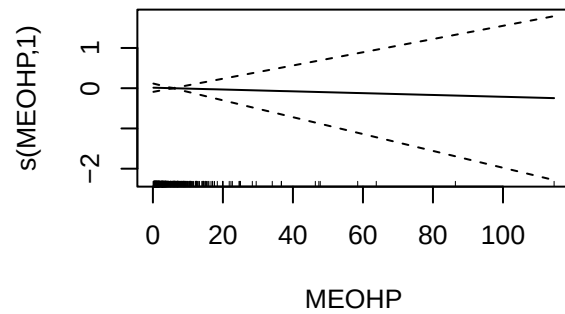
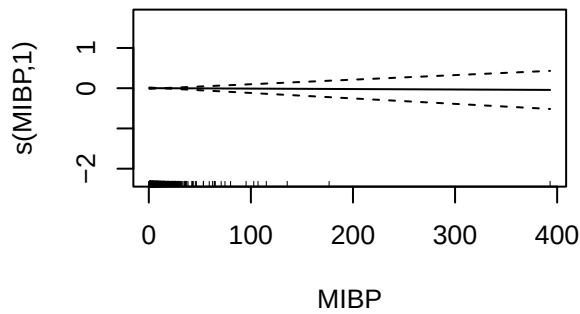
```
gam.m1 <- gam(Age_FirstPeriod ~ Age + Race + s(MCNP) + s(MCOP) + s(MECP) + s(MBP) +  
              s(MCPP) + s(MEP) + s(MEHHP) + s(MEHP) + s(MIBP) + s(MEOHP) + s(MBZP) +  
              Year,  
              data = data,  
              family = poisson)  
summary(gam.m1)
```

```
##  
## Family: poisson  
## Link function: log  
##  
## Formula:  
## Age_FirstPeriod ~ Age + Race + s(MCNP) + s(MCOP) + s(MECP) +  
##      s(MBP) + s(MCPP) + s(MEP) + s(MEHHP) + s(MEHP) + s(MIBP) +  
##      s(MEOHP) + s(MBZP) + Year  
##  
## Parametric coefficients:  
##              Estimate Std. Error z value Pr(>|z|)  
## (Intercept)      2.5038439   0.0826413  30.298   <2e-16 ***  
## Age              0.0009202   0.0014594   0.631    0.528  
## RaceNonHispanic_Black -0.0051571   0.0416891  -0.124    0.902  
## RaceNonHispanic_White -0.0329513   0.0398012  -0.828    0.408  
## RaceOther_Hispanic   -0.0458747   0.0497334  -0.922    0.356  
## RaceOtherRace_MultiRacial 0.0106291   0.0441375   0.241    0.810  
## Year2015_2016        -0.0081618   0.0311574  -0.262    0.793  
## Year2017_2018         0.0100002   0.0318397   0.314    0.753  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Approximate significance of smooth terms:  
##              edf Ref.df Chi.sq p-value  
## s(MCNP)      1.000   1.000   0.147   0.702  
## s(MCOP)      1.000   1.000   1.872   0.171  
## s(MECP)      1.000   1.000   0.011   0.916  
## s(MBP)       1.000   1.000   0.503   0.478  
## s(MCPP)      1.247   1.455   1.465   0.454  
## s(MEP)       1.000   1.000   0.346   0.557  
## s(MEHHP)     1.054   1.106   0.118   0.852  
## s(MEHP)      1.000   1.000   0.005   0.942  
## s(MIBP)      1.000   1.000   0.031   0.861  
## s(MEOHP)     1.000   1.000   0.056   0.813  
## s(MBZP)      1.000   1.000   0.111   0.739  
##  
## R-sq.(adj) =  0.0306   Deviance explained = 6.55%  
## UBRE = -0.66611   Scale est. = 1           n = 513
```

```
# Generate plots for the nonlinear terms included in the model  
plot(gam.m1, pages=4)
```







Model using outcome as Age_FirstPeriod (counts)

```
# Model Training using caret

# Matrix of predictors
x1 <- model.matrix(Age_FirstPeriod ~ Age + Race + MCNP + MCOP + MECPP + MBP + MCPP +
  MEP + MEHHP + MEHP + MIBP + MEOHP + MBZP + Year, data)[, -1]

# Vector of response
y1 <- data$Age_FirstPeriod
# Set up CV
ctrl1 <- trainControl(method = "cv", number = 10)
# Set seed for reproducibility
set.seed(2)
# Train model
gam.fit1 <- train(x1, y1,
  method = "gam",
  family = poisson,
  trControl = ctrl1)

# Best tuning parameters
gam.fit1$bestTune
# Final model
gam.fit1$finalModel
# Generate plots for the nonlinear terms included in the model
plot(gam.fit1$finalModel, pages=4)
```

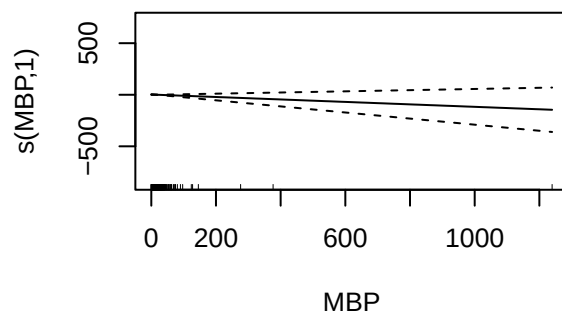
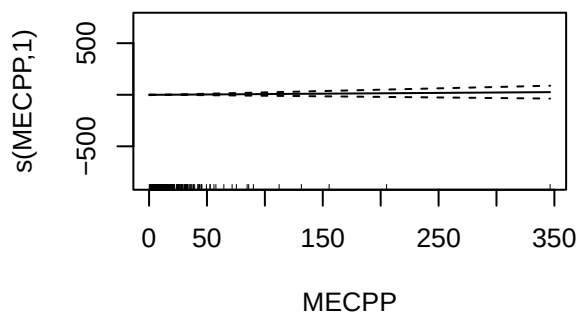
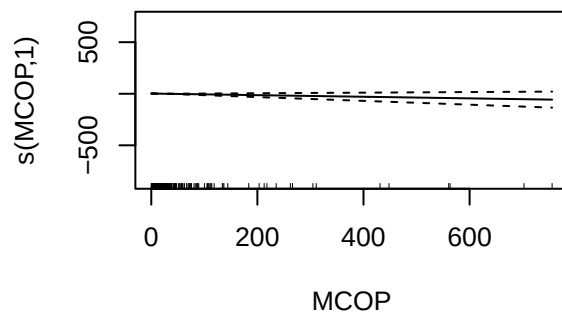
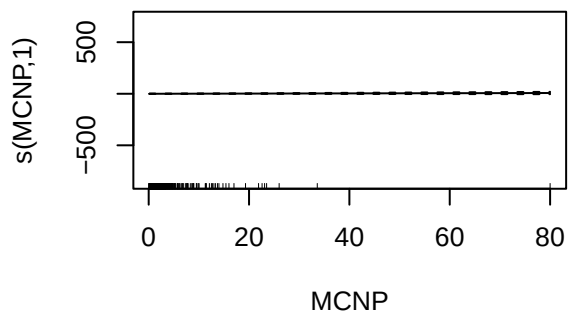
Model using outcome as RegularPeriod_12m (binary)

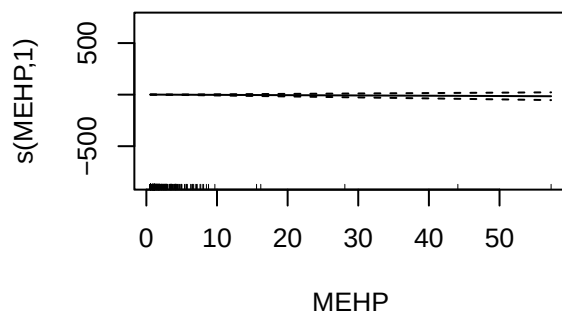
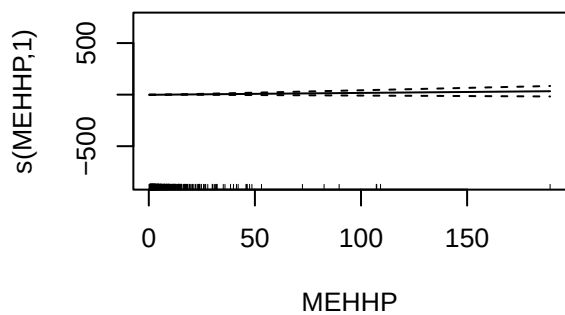
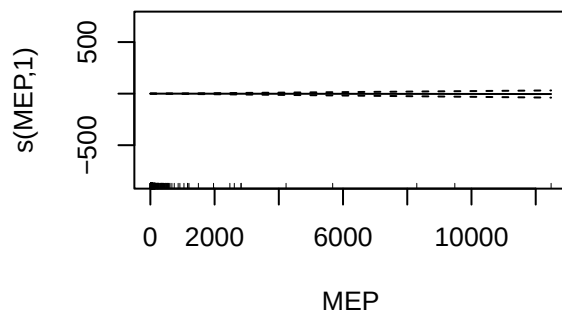
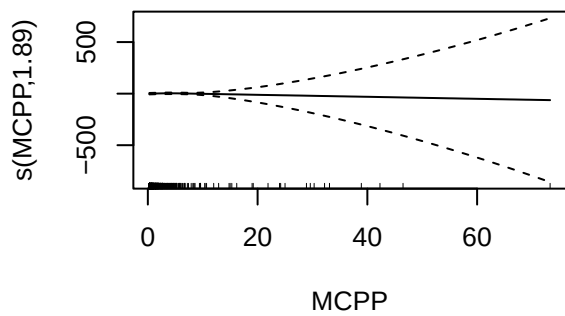
```
# Model Fitting using mgcv

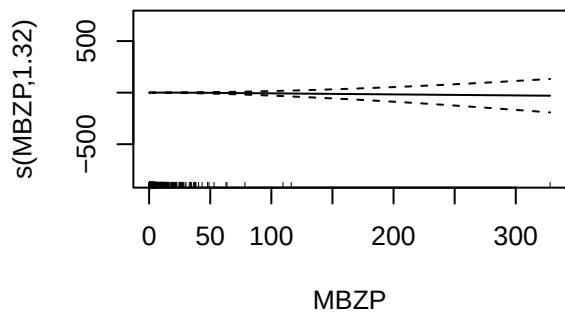
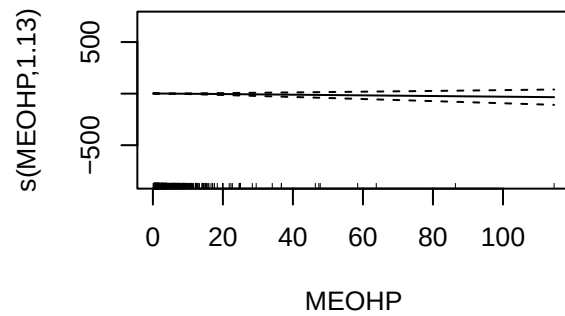
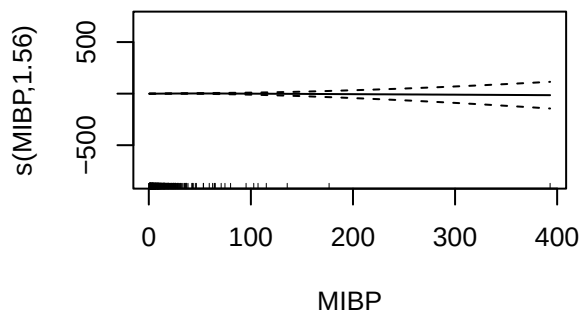
# Fit model
gam.m2 <- gam(RegularPeriod_12m ~ Age + Race + s(MCNP) + s(MCOP) + s(MECP) + s(MBP) +
              s(MCPP) + s(MEP) + s(MEHHP) + s(MEHP) + s(MIBP) + s(MEOHP) + s(MBZP) +
              Year,
              data = data,
              family = binomial)
summary(gam.m2)

##
## Family: binomial
## Link function: logit
##
## Formula:
## RegularPeriod_12m ~ Age + Race + s(MCNP) + s(MCOP) + s(MECP) +
##      s(MBP) + s(MCPP) + s(MEP) + s(MEHHP) + s(MEHP) + s(MIBP) +
##      s(MEOHP) + s(MBZP) + Year
##
## Parametric coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -6.914e+01  7.599e+06   0.000 0.999993
## Age           -1.348e-01  3.714e-02  -3.631 0.000283 ***
## RaceNonHispanic_Black  6.845e+01  7.599e+06   0.000 0.999993
## RaceNonHispanic_White  6.781e+01  7.599e+06   0.000 0.999993
## RaceOther_Hispanic     6.911e+01  7.599e+06   0.000 0.999993
## RaceOtherRace_MultiRacial 6.871e+01  7.599e+06   0.000 0.999993
## Year2015_2016          -9.912e-01  1.186e+00  -0.836 0.403245
## Year2017_2018          -2.967e-01  1.074e+00  -0.276 0.782302
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##              edf Ref.df Chi.sq p-value
## s(MCNP)      1.000  1.000  1.564  0.211
## s(MCOP)      1.000  1.000  2.056  0.152
## s(MECP)      1.000  1.000  0.636  0.425
## s(MBP)       1.000  1.000  1.758  0.185
## s(MCPP)      1.891  2.218  3.128  0.172
## s(MEP)       1.000  1.000  0.046  0.830
## s(MEHHP)     1.000  1.000  1.633  0.201
## s(MEHP)      1.000  1.000  0.648  0.421
## s(MIBP)      1.560  1.909  1.295  0.547
## s(MEOHP)     1.130  1.232  0.611  0.448
## s(MBZP)      1.324  1.560  0.244  0.740
##
## R-sq.(adj) =  0.27   Deviance explained = 40.3%
## UBRE = -0.81303   Scale est. = 1           n = 513
```

```
# Generate plots for the nonlinear terms included in the model
plot(gam.m2, pages=4)
```







Model using outcome as RegularPeriod_12m (binary)

```
# Model Training using caret

# Matrix of predictors
x2 <- model.matrix(RegularPeriod_12m ~ Age + Race + MCNP + MCOP + MECPP + MBP + MCPP +
                    MEP + MEHHP + MEHP + MIBP + MEOHP + MBZP + Year, data)[, -1]

# Vector of response
y2 <- data$RegularPeriod_12m
# Set seed for reproducibility
set.seed(2)
# Train model
gam.fit2 <- train(x2, y2,
                  method = "gam",
                  family = binomial,
                  trControl = ctrl1)

# Best tuning parameters
gam.fit2$bestTune
# Final model
gam.fit2$finalModel
# Generate plots for the nonlinear terms included in the model
plot(gam.fit2$finalModel, pages=4)
```

Model using outcome as UnablePregnant (binary)

```
# Model Fitting using mgcv
```

```
# Fit model
```

```
gam.m3 <- gam(UnablePregnant ~ Age + Race + s(MCNP) + s(MCOP) + s(MECP) + s(MBP) +  
              s(MCPP) + s(MEP) + s(MEHP) + s(MEHP) + s(MIBP) + s(MEOHP) + s(MBZP) +  
              Year,  
              data = data,  
              family = binomial)
```

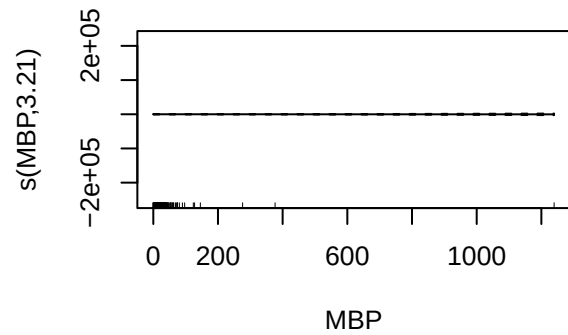
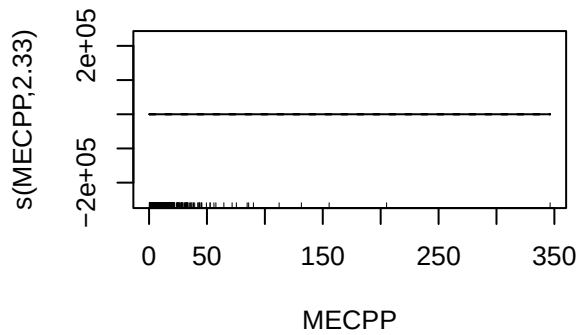
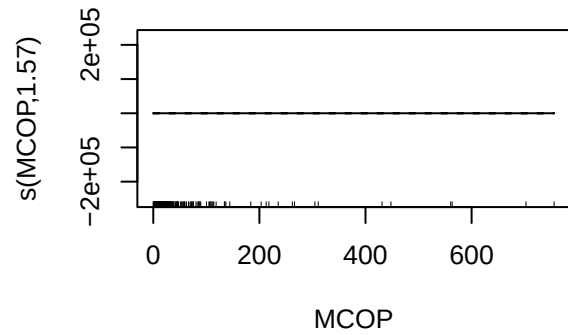
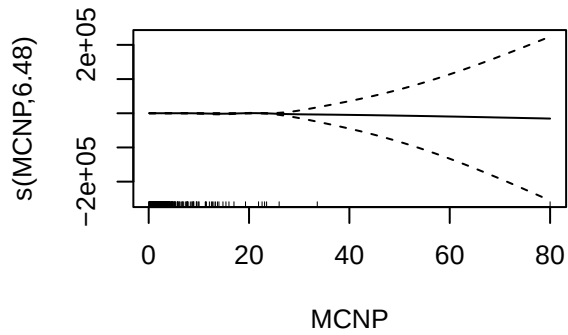
```
## Warning in newton(lsp = lsp, X = G$X, y = G$y, Eb = G$Eb, UrS = G$UrS, L = G$L,  
## : Iteration limit reached without full convergence - check carefully
```

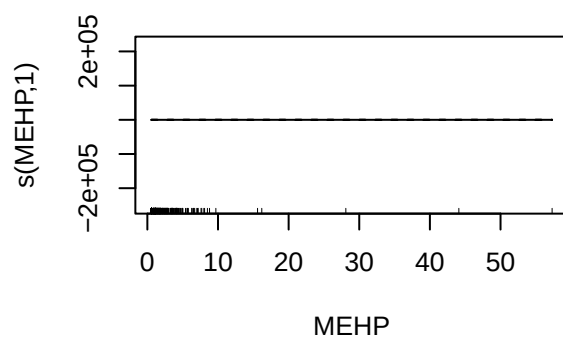
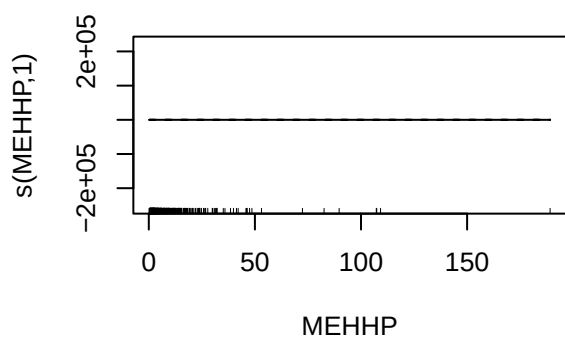
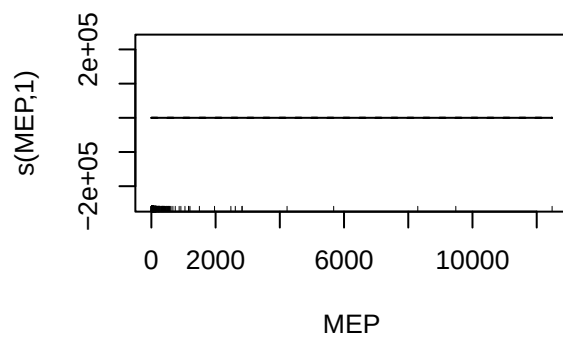
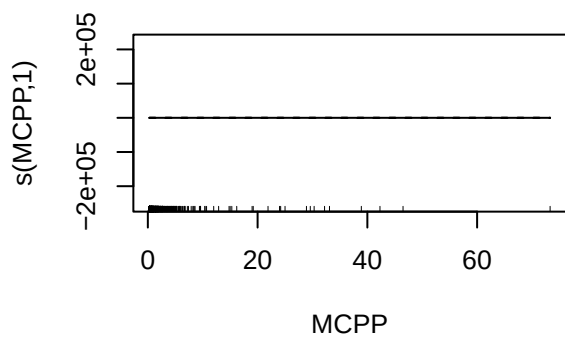
```
summary(gam.m3)
```

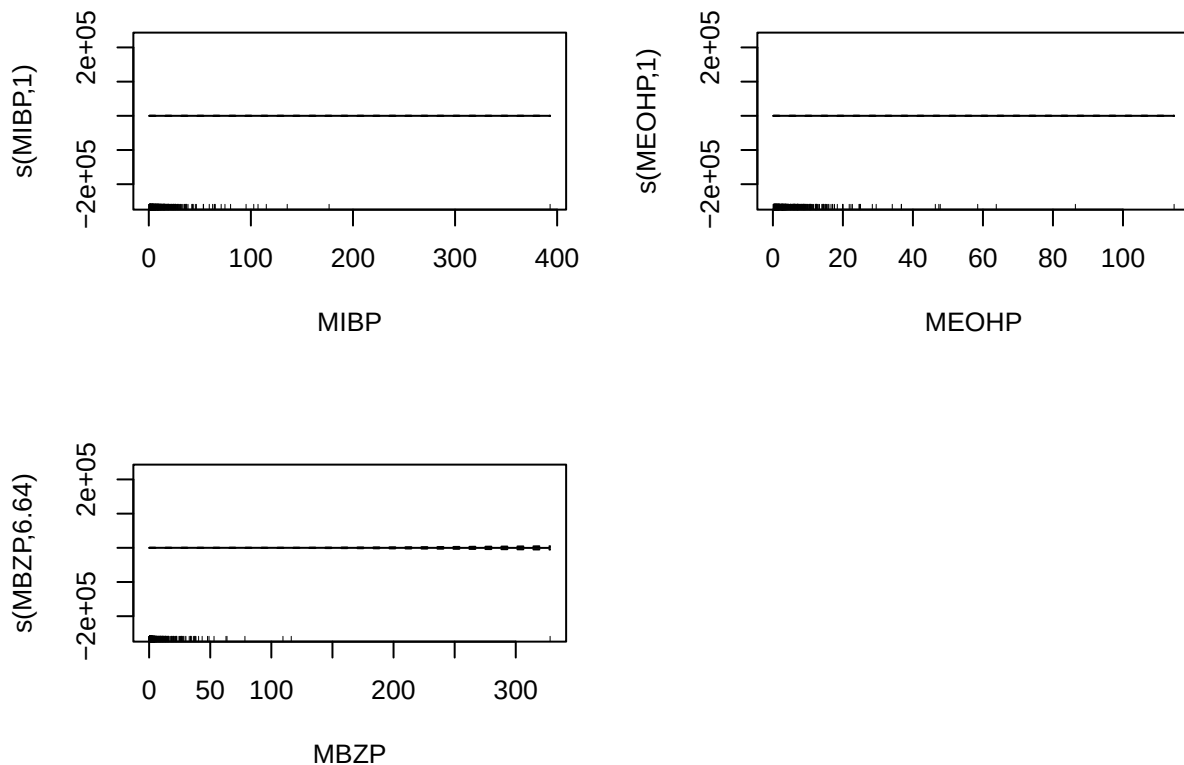
```
##  
## Family: binomial  
## Link function: logit  
##  
## Formula:  
## UnablePregnant ~ Age + Race + s(MCNP) + s(MCOP) + s(MECP) +  
##      s(MBP) + s(MCPP) + s(MEP) + s(MEHP) + s(MEHP) + s(MIBP) +  
##      s(MEOHP) + s(MBZP) + Year  
##  
## Parametric coefficients:  
##              Estimate Std. Error z value Pr(>|z|)  
## (Intercept)    -87.61561   258.64433  -0.339   0.7348  
## Age              0.04320    0.02501   1.727   0.0842 .  
## RaceNonHispanic_Black  1.58233    0.65824   2.404   0.0162 *  
## RaceNonHispanic_White  1.19291    0.66064   1.806   0.0710 .  
## RaceOther_Hispanic     0.37795    0.79320   0.476   0.6337  
## RaceOtherRace_MultiRacial 0.98160    0.67896   1.446   0.1483  
## Year2015_2016          0.53329    0.42480   1.255   0.2093  
## Year2017_2018          0.34498    0.46300   0.745   0.4562  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Approximate significance of smooth terms:  
##              edf Ref.df Chi.sq p-value  
## s(MCNP)      6.482  6.737  3.133  0.858  
## s(MCOP)      1.568  1.932  1.097  0.581  
## s(MECP)      2.332  2.832  4.114  0.143  
## s(MBP)       3.214  3.787  4.170  0.251  
## s(MCPP)      1.000  1.000  1.778  0.182  
## s(MEP)       1.000  1.000  0.442  0.506  
## s(MEHP)      1.000  1.000  0.311  0.577  
## s(MEHP)      1.000  1.000  0.231  0.631  
## s(MIBP)      1.000  1.000  0.111  0.739  
## s(MEOHP)     1.000  1.000  0.054  0.816  
## s(MBZP)      6.636  6.934  9.739  0.196  
##
```

```
## R-sq.(adj) = 0.122   Deviance explained = 23.2%
## UBRE = -0.36908   Scale est. = 1           n = 513
```

```
# Generate plots for the nonlinear terms included in the model
plot(gam.m3, pages=4)
```







Model using outcome as UnablePregnant (binary)

```
# Model Training using caret

# Matrix of predictors
x3 <- model.matrix(UnablePregnant ~ Age + Race + MCNP + MCOP + MECPP + MBP + MCPP +
                    MEP + MEHHP + MEHP + MIBP + MEOHP + MBZP + Year, data)[, -1]

# Vector of response
y3 <- data$UnablePregnant
# Set seed for reproducibility
set.seed(2)
# Train model
gam.fit3 <- train(x3, y3,
                  method = "gam",
                  family = binomial,
                  trControl = ctrl1)

# Best tuning parameters
gam.fit3$bestTune
# Final model
gam.fit3$finalModel
# Generate plots for the nonlinear terms included in the model
plot(gam.fit3$finalModel, pages=4)
```

Participants under 50

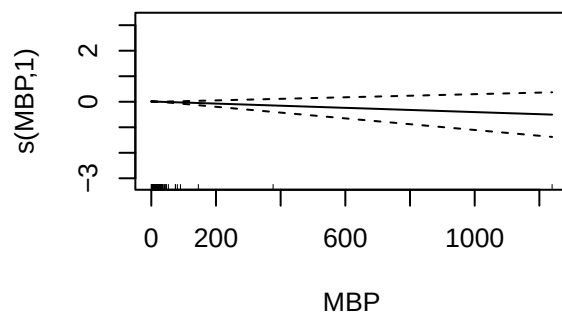
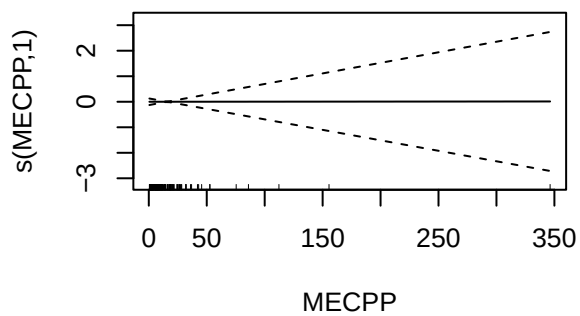
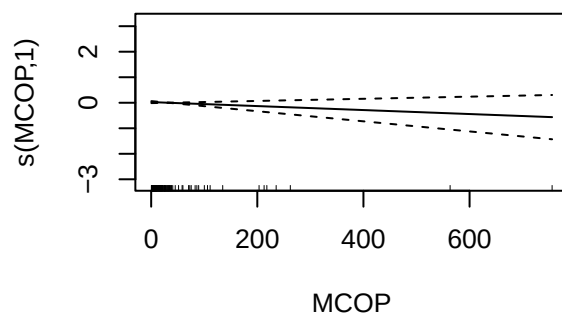
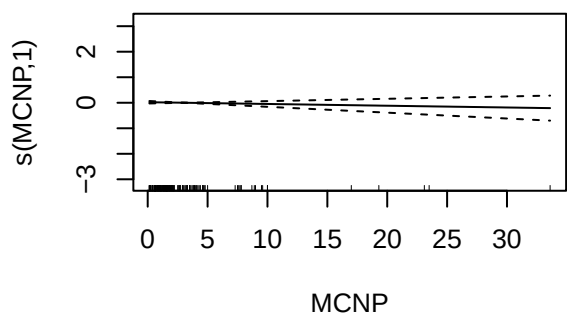
Model using outcome as Age_FirstPeriod (counts)

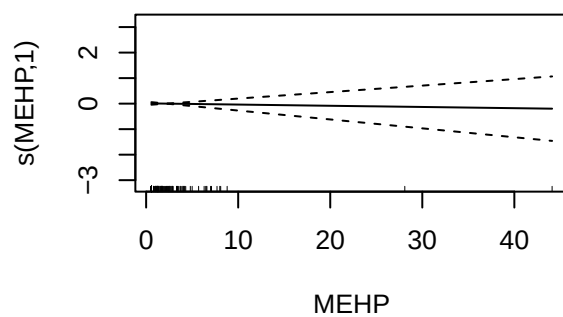
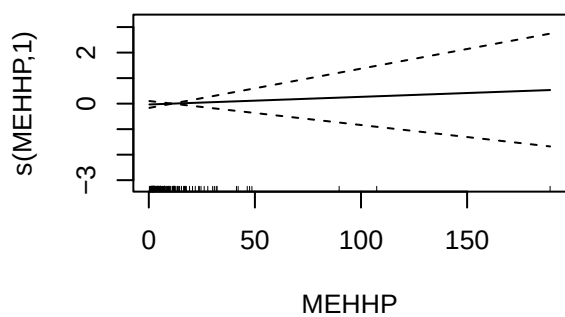
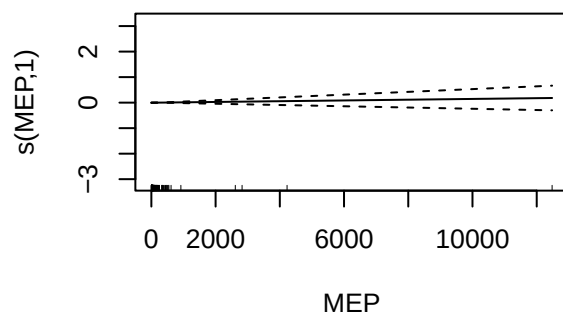
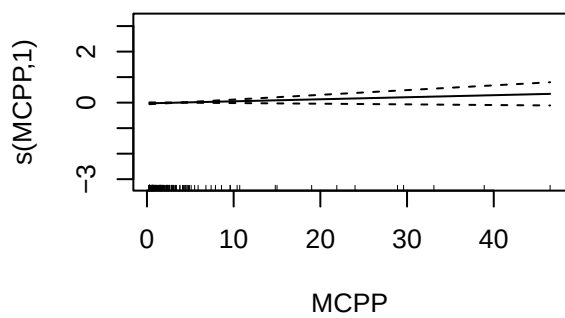
```
# Model Fitting using mgcv

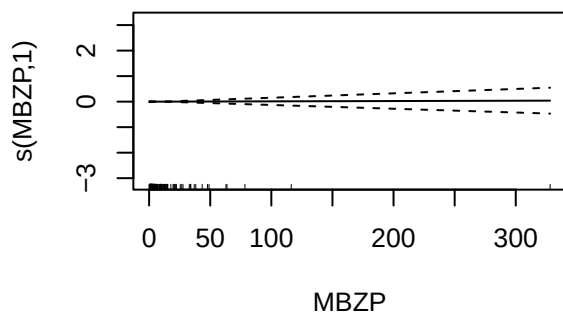
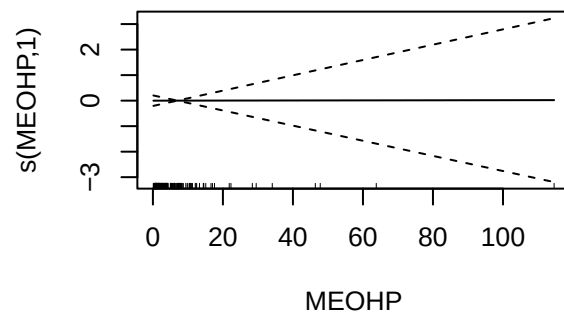
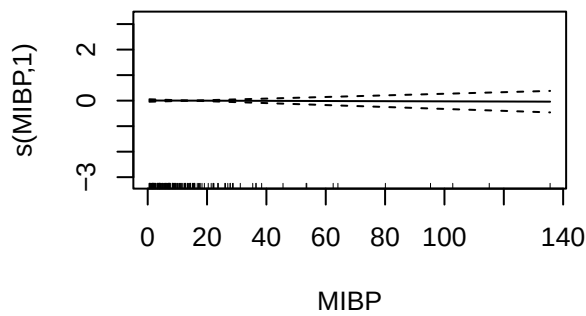
# Fit model
gam.m1_under50 <- gam(Age_FirstPeriod ~ Age + Race + s(MCNP) + s(MCOP) + s(MECP) + s(MBP) +
                      s(MCPP) + s(MEP) + s(MEHHP) + s(MEHP) + s(MIBP) + s(MEOHP) + s(MBZP) +
                      Year,
                      data = data_under50,
                      family = poisson)
summary(gam.m1_under50)

##
## Family: poisson
## Link function: log
##
## Formula:
## Age_FirstPeriod ~ Age + Race + s(MCNP) + s(MCOP) + s(MECP) +
##      s(MBP) + s(MCPP) + s(MEP) + s(MEHHP) + s(MEHP) + s(MIBP) +
##      s(MEOHP) + s(MBZP) + Year
##
## Parametric coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    2.5074592   0.1295190   19.360   <2e-16 ***
## Age            0.0004944   0.0028021    0.176    0.860
## RaceNonHispanic_Black  0.0235433   0.0787481    0.299    0.765
## RaceNonHispanic_White -0.0390754   0.0684389   -0.571    0.568
## RaceOther_Hispanic    0.0123041   0.1011137    0.122    0.903
## RaceOtherRace_MultiRacial -0.0218576   0.0836443   -0.261    0.794
## Year2015_2016    0.0069879   0.0574036    0.122    0.903
## Year2017_2018    0.0166128   0.0636232    0.261    0.794
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##              edf Ref.df Chi.sq p-value
## s(MCNP)      1      1  0.708  0.400
## s(MCOP)      1      1  1.621  0.203
## s(MECP)      1      1  0.000  0.994
## s(MBP)       1      1  1.278  0.258
## s(MCPP)      1      1  2.195  0.138
## s(MEP)       1      1  0.542  0.462
## s(MEHHP)     1      1  0.223  0.637
## s(MEHP)      1      1  0.094  0.759
## s(MIBP)      1      1  0.035  0.852
## s(MEOHP)     1      1  0.000  0.991
## s(MBZP)      1      1  0.022  0.881
##
## R-sq.(adj) =  0.104   Deviance explained = 20.3%
## UBRE = -0.54602   Scale est. = 1          n = 158
```

```
# Generate plots for the nonlinear terms included in the model
plot(gam.m1_under50, pages=4)
```







Model using outcome as Age_FirstPeriod (counts)

```
# Model Training using caret

# Matrix of predictors
x1_under50 <- model.matrix(Age_FirstPeriod ~ Age + Race + MCNP + MCOP + MECPP + MBP + MCPP +
                           MEP + MEHHP + MEHP + MIBP + MEOHP + MBZP + Year, data_under50)[, -1]

# Vector of response
y1_under50 <- data_under50$Age_FirstPeriod

# Set seed for reproducibility
set.seed(2)

# Train model
gam.fit1_under50 <- train(x1_under50, y1_under50,
                          method = "gam",
                          family = poisson,
                          trControl = ctrl1)

# Best tuning parameters
gam.fit1_under50$bestTune

# Final model
gam.fit1_under50$finalModel

# Generate plots for the nonlinear terms included in the model
plot(gam.fit1_under50$finalModel, pages=4)
```

Model using outcome as RegularPeriod_12m (binary)

```
# Model Fitting using mgcv
```

```
# Fit model
```

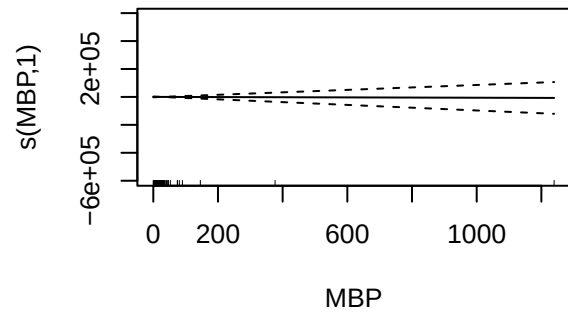
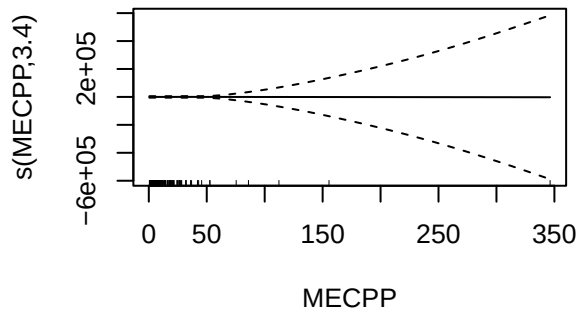
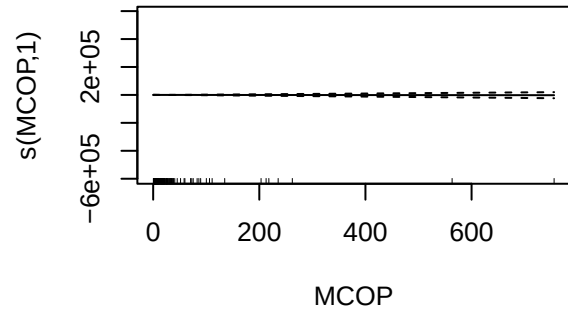
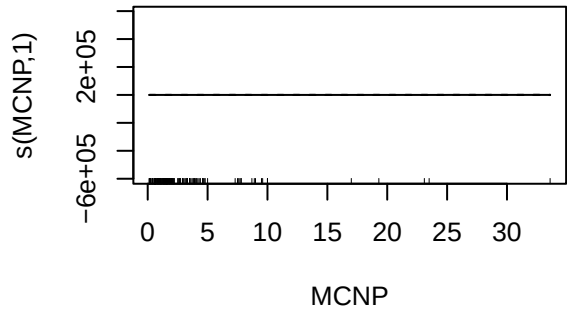
```
gam.m2_under50 <- gam(RegularPeriod_12m ~ Age + Race + s(MCNP) + s(MCOP) + s(MECP) + s(MBP) +  
  s(MCPP) + s(MEP) + s(MEHHP) + s(MEHP) + s(MIBP) + s(MEOHP) + s(MBZP) +  
  Year,  
  data = data_under50,  
  family = binomial)
```

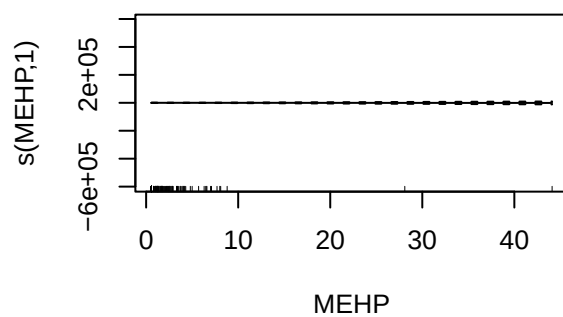
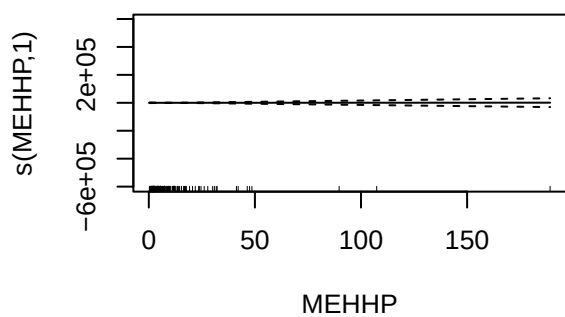
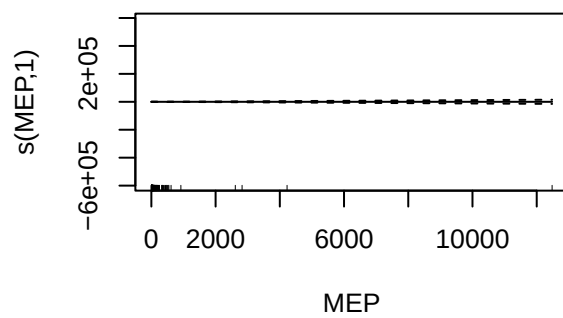
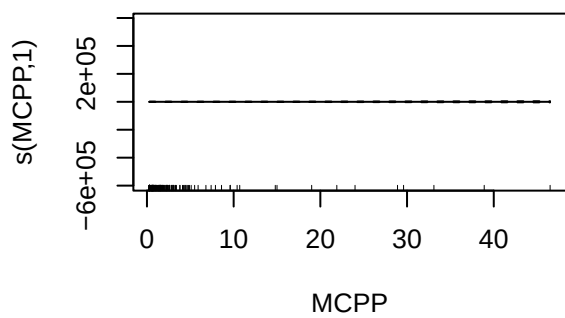
```
## Warning in newton(lsp = lsp, X = G$X, y = G$y, Eb = G$Eb, UrS = G$UrS, L = G$L,  
## : Iteration limit reached without full convergence - check carefully
```

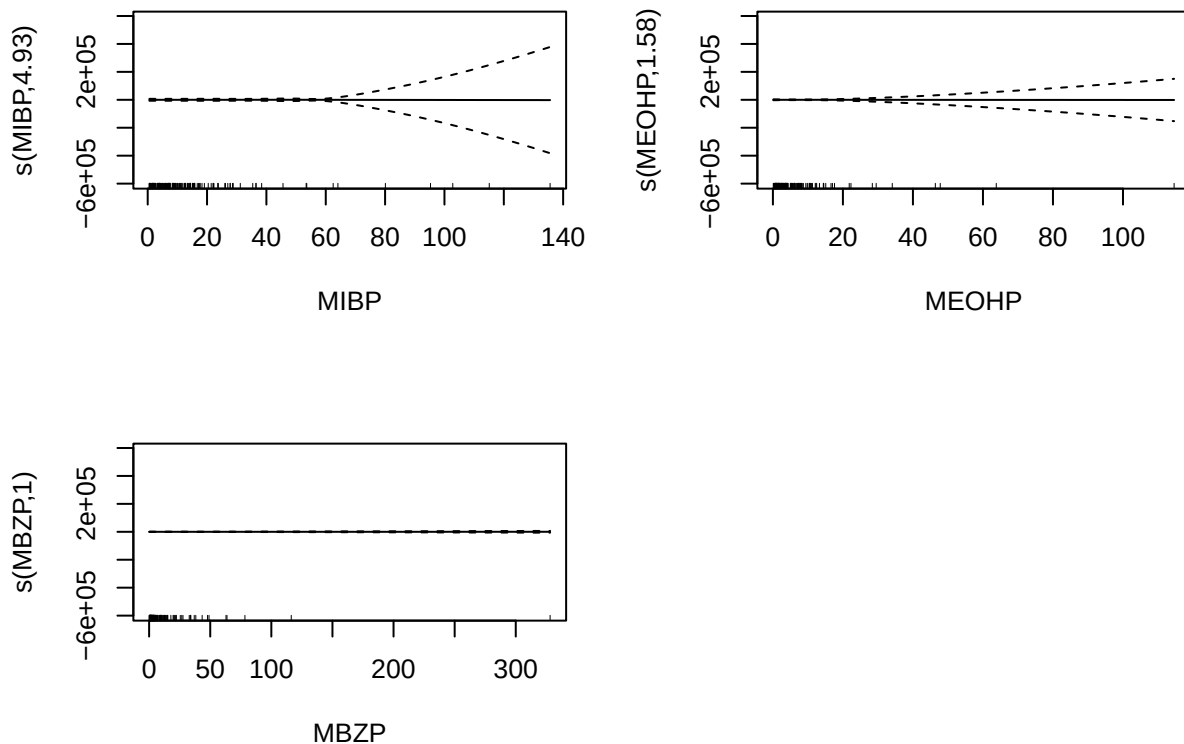
```
summary(gam.m2_under50)
```

```
##  
## Family: binomial  
## Link function: logit  
##  
## Formula:  
## RegularPeriod_12m ~ Age + Race + s(MCNP) + s(MCOP) + s(MECP) +  
##   s(MBP) + s(MCPP) + s(MEP) + s(MEHHP) + s(MEHP) + s(MIBP) +  
##   s(MEOHP) + s(MBZP) + Year  
##  
## Parametric coefficients:  
##              Estimate Std. Error z value Pr(>|z|)  
## (Intercept)    -6.890e+02  1.292e+07   0.000   1.000  
## Age            -2.010e-01  5.190e+00  -0.039   0.969  
## RaceNonHispanic_Black  4.605e+02  1.292e+07   0.000   1.000  
## RaceNonHispanic_White  4.142e+02  1.292e+07   0.000   1.000  
## RaceOther_Hispanic    4.323e+02  1.292e+07   0.000   1.000  
## RaceOtherRace_MultiRacial 4.389e+02  1.292e+07   0.000   1.000  
## Year2015_2016      -1.019e+01  2.206e+02  -0.046   0.963  
## Year2017_2018      -5.625e+00  2.068e+02  -0.027   0.978  
##  
## Approximate significance of smooth terms:  
##              edf Ref.df Chi.sq p-value  
## s(MCNP)      1.000  1.000  0.055  0.815  
## s(MCOP)      1.000  1.000  0.031  0.861  
## s(MECP)      3.395  3.506  0.013  1.000  
## s(MBP)       1.000  1.000  0.015  0.902  
## s(MCPP)      1.000  1.000  0.000  0.984  
## s(MEP)       1.000  1.000  0.000  0.998  
## s(MEHHP)     1.000  1.000  0.004  0.951  
## s(MEHP)      1.000  1.000  0.031  0.861  
## s(MIBP)      4.932  5.006  0.065  1.000  
## s(MEOHP)     1.579  1.670  0.005  0.996  
## s(MBZP)      1.000  1.000  0.001  0.976  
##  
## R-sq.(adj) =      1   Deviance explained = 100%  
## UBRE = -0.67203  Scale est. = 1           n = 158
```

```
# Generate plots for the nonlinear terms included in the model
plot(gam.m2_under50, pages=4)
```







Model using outcome as RegularPeriod_12m (binary)

```
# Model Training using caret

# Matrix of predictors
x2_under50 <- model.matrix(RegularPeriod_12m ~ Age + Race + MCNP + MCOP + MECPP + MBP + MCPP +
                           MEP + MEHHP + MEHP + MIBP + MEOHP + MBZP + Year, data_under50)[, -1]

# Vector of response
y2_under50 <- data_under50$RegularPeriod_12m
# Set seed for reproducibility
set.seed(2)
# Train model
gam.fit2_under50 <- train(x2_under50, y2_under50,
                          method = "gam",
                          family = binomial,
                          trControl = ctrl1)

# Best tuning parameters
gam.fit2_under50$bestTune
# Final model
gam.fit2_under50$finalModel
# Generate plots for the nonlinear terms included in the model
plot(gam.fit2_under50$finalModel, pages=4)
```

Model using outcome as UnablePregnant (binary)

```
# Model Fitting using mgcv
```

```
# Fit model
```

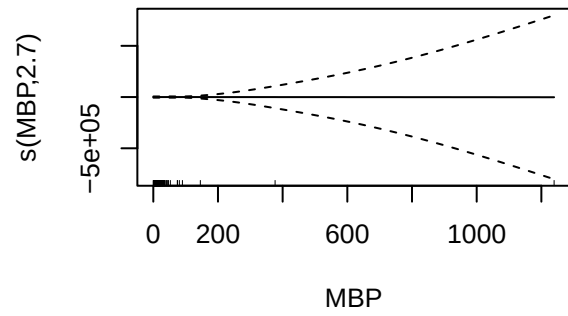
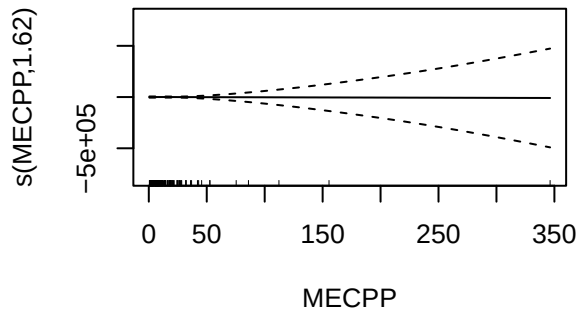
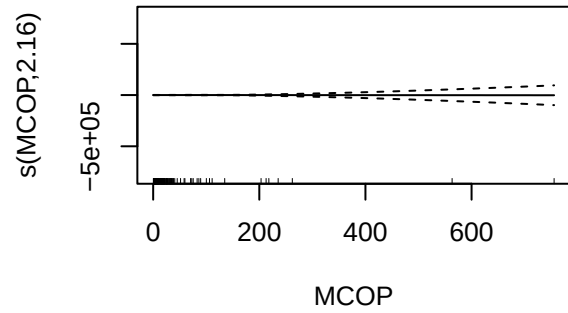
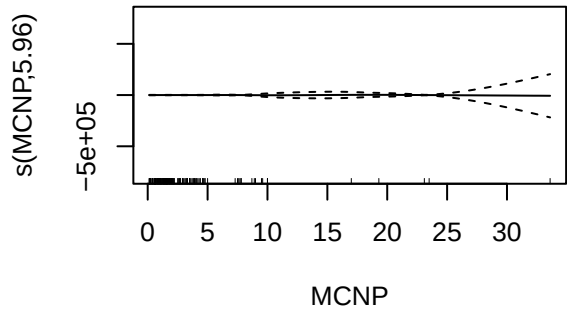
```
gam.m3_under50 <- gam(UnablePregnant ~ Age + Race + s(MCNP) + s(MCOP) + s(MECP) + s(MBP) +  
                      s(MCPP) + s(MEP) + s(MEHHP) + s(MEHP) + s(MIBP) + s(MEOHP) + s(MBZP) +  
                      Year,  
                      data = data_under50,  
                      family = binomial)
```

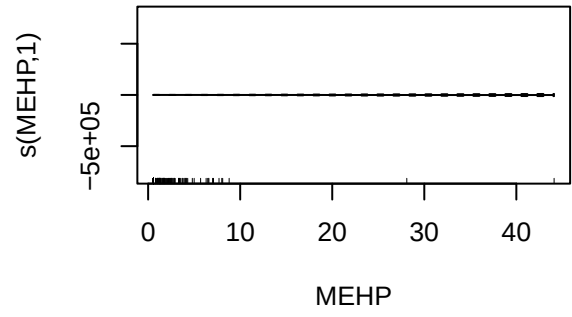
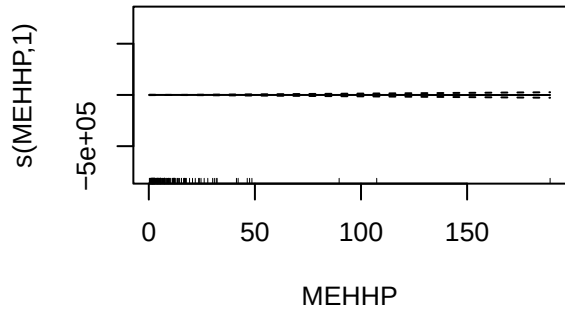
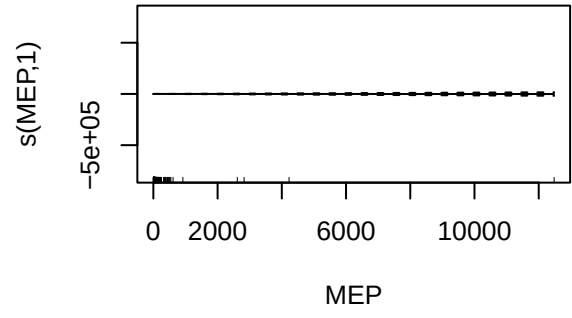
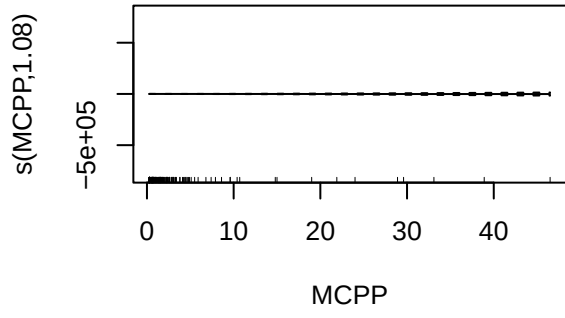
```
## Warning in newton(lsp = lsp, X = G$X, y = G$y, Eb = G$Eb, UrS = G$UrS, L = G$L,  
## : Iteration limit reached without full convergence - check carefully
```

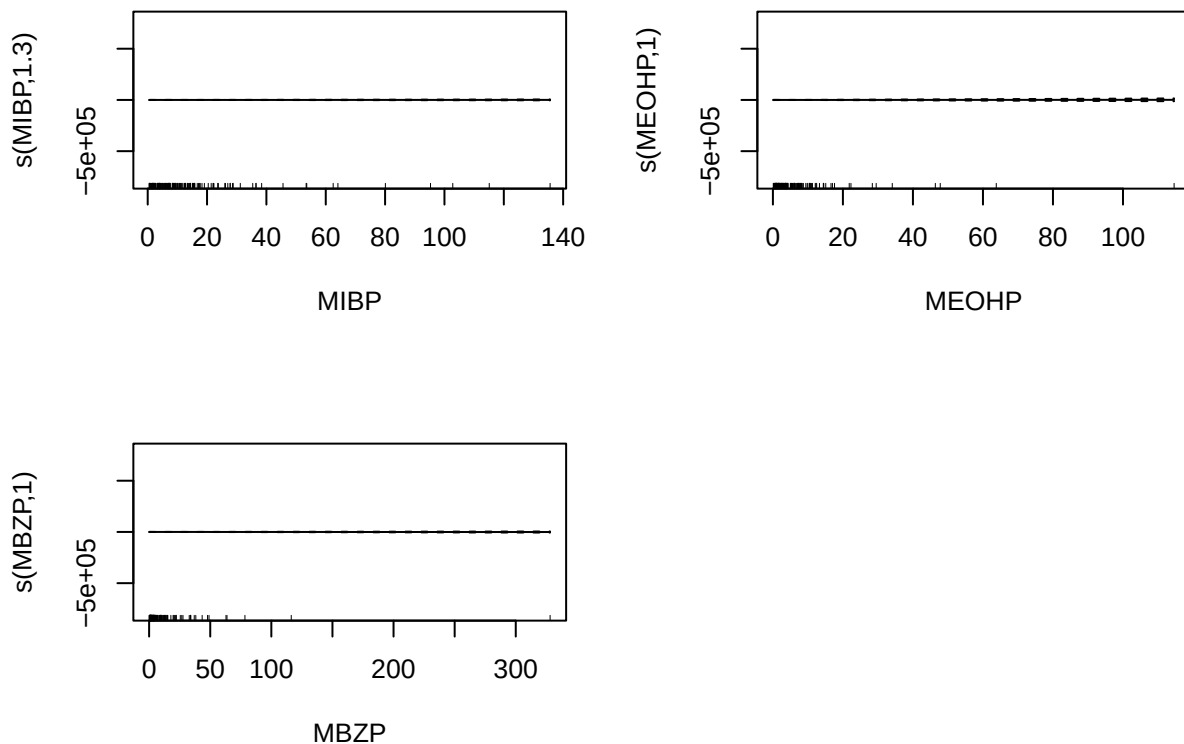
```
summary(gam.m3_under50)
```

```
##  
## Family: binomial  
## Link function: logit  
##  
## Formula:  
## UnablePregnant ~ Age + Race + s(MCNP) + s(MCOP) + s(MECP) +  
##      s(MBP) + s(MCPP) + s(MEP) + s(MEHHP) + s(MEHP) + s(MIBP) +  
##      s(MEOHP) + s(MBZP) + Year  
##  
## Parametric coefficients:  
##              Estimate Std. Error z value Pr(>|z|)  
## (Intercept)    -1.143e+03  1.292e+07  0.000    1.000  
## Age             1.042e+00  8.504e+00  0.123    0.902  
## RaceNonHispanic_Black  6.780e+02  1.292e+07  0.000    1.000  
## RaceNonHispanic_White  6.597e+02  1.292e+07  0.000    1.000  
## RaceOtherHispanic    -2.088e+02  2.328e+07  0.000    1.000  
## RaceOtherRace_MultiRacial  6.677e+02  1.292e+07  0.000    1.000  
## Year2015_2016        3.507e+01  2.785e+02  0.126    0.900  
## Year2017_2018        4.635e+01  3.410e+02  0.136    0.892  
##  
## Approximate significance of smooth terms:  
##              edf Ref.df Chi.sq p-value  
## s(MCNP)      5.962  5.991  0.136    1.000  
## s(MCOP)      2.156  2.209  0.065    0.985  
## s(MECP)      1.622  1.656  0.005    0.996  
## s(MBP)       2.698  2.743  0.245    0.975  
## s(MCPP)      1.084  1.089  0.084    0.884  
## s(MEP)       1.000  1.000  0.006    0.941  
## s(MEHHP)     1.000  1.000  0.001    0.970  
## s(MEHP)      1.000  1.000  0.026    0.872  
## s(MIBP)      1.302  1.355  0.054    0.906  
## s(MEOHP)     1.000  1.000  0.030    0.862  
## s(MBZP)      1.000  1.000  0.086    0.770  
##  
## R-sq.(adj) =      1    Deviance explained = 99.8%  
## UBRE = -0.64669    Scale est. = 1          n = 158
```

```
# Generate plots for the nonlinear terms included in the model
plot(gam.m3_under50, pages=4)
```







Model using outcome as UnablePregnant (binary)

```
# Model Training using caret

# Matrix of predictors
x3_under50 <- model.matrix(UnablePregnant ~ Age + Race + MCNP + MCOP + MECP + MBP + MCPP +
                             MEP + MEHP + MEHP + MIBP + MEOHP + MBZP + Year, data_under50)[, -1]

# Vector of response
y3_under50 <- data_under50$UnablePregnant
# Set seed for reproducibility
set.seed(2)
# Train model
gam.fit3_under50 <- train(x3_under50, y3_under50,
                          method = "gam",
                          family = binomial,
                          trControl = ctrl1)

# Best tuning parameters
gam.fit3_under50$bestTune
# Final model
gam.fit3_under50$finalModel
# Generate plots for the nonlinear terms included in the model
plot(gam.fit3_under50$finalModel, pages=4)
```